

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1 Objective type questions. (Compulsory Question)			12
1	Which logic gate outputs HIGH only when all inputs are HIGH?		1
	A) OR		
	B) AND		
	C) NAND		
	D) NOR		
2	Which code is used for error detection and correction?		1
	A) Gray code		
	B) Excess-3 code		
	C) Hamming code		
	D) ASCII		
3	K-map is used to:		1
	A) Build microprocessors		
	B) Generate flip-flops		
	C) Simplify Boolean expressions		
	D) Store digital data		
4	A 4-to-16 decoder has how many inputs and outputs?		1
	A) 4 inputs, 16 outputs		
	B) 16 inputs, 4 outputs		
	C) 2 inputs, 8 outputs		
	D) 8 inputs, 2 outputs		
5	The basic memory element in sequential logic is:		1
	A) Multiplexer		
	B) Decoder		
	C) Flip-Flop		
	D) Encoder		
6	JK flip-flop avoids race condition by:		1
	A) Using only one input		
	B) Feeding output back to input		
	C) Using a clock pulse		
	D) Avoiding logic gates		

- 7 What is the main function of a shift register?
 - A) Arithmetic calculation
 - B) Storage
 - C) Data conversion
 - D) Data shifting
- 8 The ALU in a microprocessor performs:
 - A) Only addition
 - B) Arithmetic and logical operations
 - C) Memory access
 - D) Clock generation
- 9 How many address lines are used by the 8086 microprocessor?
 - A) 8
 - B) 16
 - C) 20
 - D) 32
- 10 DMA stands for:
 - A) Direct Machine Access
 - B) Direct Memory Access
 - C) Digital Memory Address
 - D) Device Management Architecture
- 11 Which addressing mode uses a constant as an operand?
 - A) Direct
 - B) Indirect
 - C) Immediate
 - D) Indexed
- 12 The function of the Program Counter (PC) is to:
 - A) Store data
 - B) Store result of ALU
 - C) Hold the address of next instruction
 - D) Point to the stack

Q.2 Solve the following.

- A) Convert the following (L2)
 - a) $(1AD)_{16} = ()_{10}$
 - b) $(453)_8 = ()_{10}$
 - c) $(10110011)_2 = ()_{10}$
- B) Design a 4:1 multiplexer and explain its working with a logic diagram and truth table. (L3)

Q.3 Solve the following.

- A) Design the circuit by Using NAND gates $F = ABC' + DE + AB'D'$ (L3)
- B) Perform the following operations using 2's complement method (L3)
 - (i) $(1101)_2 + (1001)_2$,
 - (ii) $(1010)_2 - (1101)_2$

Q. 4 Solve Any Two of the following.

- A) Design a 3 bit odd parity generator and implement with NAND gates. (L6)
- B) Differentiate between Synchronous & Asynchronous counters. Draw and explain 4-bit Ripple counter with waveforms (L4)
- C) Simplify the Boolean expression using K-MAP
 $F(A,B,C,D) = \sum m (1,2,3,8,9,10,11,14) + d (7,15)$ (L4)

Q.5 Solve Any Two of the following.

- A) Draw the architecture of 8086 and explain the function of ALU, Bus Interface Unit, and Execution Unit. (L2)
- B) Design a 3-bit asynchronous counter using JK flip-flops. Explain its timing diagram. (L3)
- C) Explain the working of a Serial-In Parallel-Out shift register. (L2)

Q. 6 Solve Any Two of the following. <https://www.batuonline.com>

- A) Write an 8086 assembly language program to add two 8-bit numbers stored at memory locations 3000H and 3001H. Store the result in 3002H. (L3)
- B) Explain the different addressing modes in 8086 with suitable examples. (L2)
- C) Draw and explain the working of a clocked SR flip-flop. Compare it with D flip-flop. (L4)

***** End *****